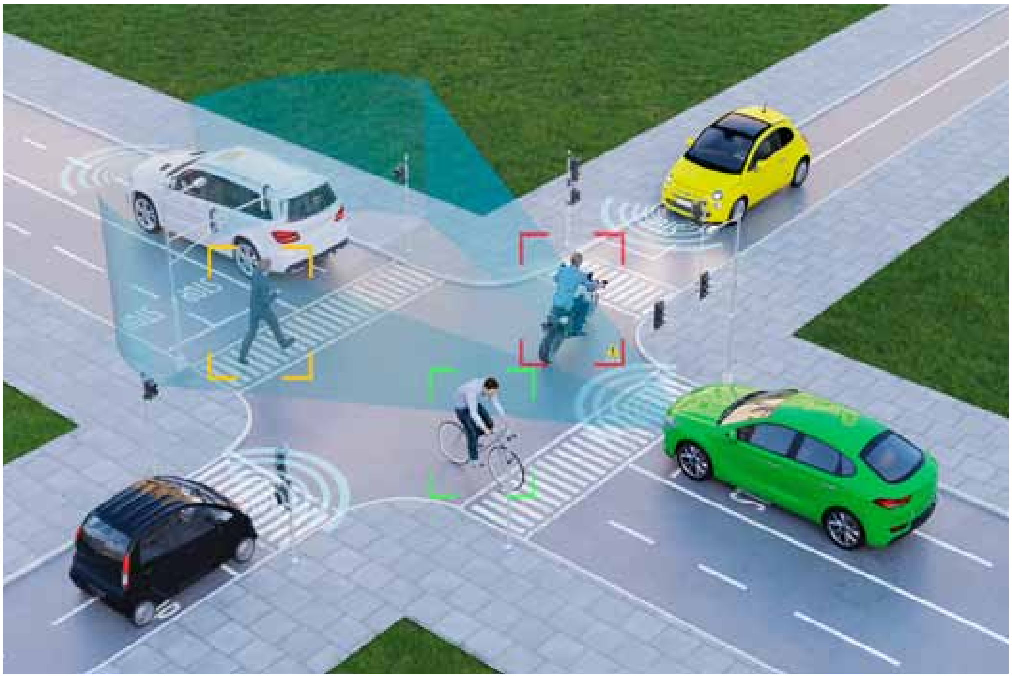

The Internet of Things (IoT) would be crucial for vehicle connectivity. The connected vehicles capable of communicating with the surrounding environment will lead to the emergence of new-age business opportunities through connected data. These enabling technologies for futuristic vehicles will be commercially available in the next 4-5 years.

The two-wheeler segment will also witness increased adoption of electronics in the future. The three-wheeler segment is a laggard in the adoption of electronics. Embedded telematics and infotainment offerings are occasional in the three-wheelers category and are expected to remain the same over the short to medium term.

Domestic capacity building

Domestic capacity building would be a key to tapping this potential. Indian suppliers have built a strong capability in hardware manufacturing, forging, and machining of the components but are comparatively new in electronics design and manufacturing. As a result, majority of the demand for automotive electronics is met through imports, primarily from China, Taiwan, and Korea.

The top five products, namely, engine control unit (ECU), EV/HV, HVAC, infotainment, and lighting, account for 95% of the automotive electronics demand in the country, according to Frost & Sullivan's analysis. Government initiatives, such as the Automotive Mission Plan, which aims to produce 940 million vehicles by FY26, are expected to generate significant demand for automotive electronics over the next five years.

The National Policy on Electronics (NPE 2019) aims to position India as a global hub for electronics system design and manufacturing (ESDM) by encouraging and driving country capabilities to develop core components, including chipsets, and create an enabling environment for the industry to compete globally. The NPE 2019 also envisions the creation of a vibrant and dynamic semiconductor design ecosystem in the country by incentivising the start-ups and making design infrastructure accessible to them.

Towards this, the government has promoted the entire ecosystem of the Indian electronics industry through incentive schemes. Incentive support is being provided to companies/consortia that are engaged in silicon semiconductor fabs, display fabs and compound semiconductors/silicon photonics/sensors (including MEMS), fabs semiconductor packaging (ATMP/OSAT) and semiconductor design (design linked incentive or DLI).

PLI for IT hardware and large scale electronics

Production linked incentive (PLI) schemes will boost investment in the entire value chain of the Indian electronics industry, including designing, ensuring local availability of components (ICs, chipsets, systems on chip, systems or IP cores, etc), and make Indian electronics industry more self-reliant and export-oriented. Developing the local manufacturing ecosystem will strengthen the local supply chain, thereby improving time to market, reducing lead times, saving precious foreign exchange, reducing component and logistics costs, and making electronics products more affordable in the coming years.

Electronics manufacturers can now focus on improving capabilities in such areas as PCB manufacturing techniques, SMT based automated production lines, ESD precautions in handling electronic components, in-circuit testing procedures, and PLC based automated techniques, to name a few. Additionally, Indian manufacturers need to build superior capability in hardware design, testing, software integration, in-vehicle networking, and Simulink tools for creating and simulating interface systems. EFY